AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM270
Artificial intelligence

الجامعة العربية المفتوحة - مصر
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القبول والتسجيل
الامتحانات
Arab Republic of Egypt

Midterm Examination (MTA)
Spring Semester 2025/2026

Date: 21 /April/ 2026

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 120 minutes

Instructions:

- Total Marks: **60**
- This exam consists of **2** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Part -I: Question 1: Choose the best correct answer of all the following: (15 Marks:1.5 per question)

1 ..... can apply their effectors further afield than anchored manipulators can, but their task is made harder because they don't have the rigidity that the anchor provides.

A Manipulators
B mobile manipulator
C Mobile robot
D None of the above

2 In the ..... search, given a state, we only know whether it is a goal state or not

A uninformed
B adaptive
C smart
D informed

3 Stone game is an example of .....games.

A stochastic
B random
C deterministic
D None of the above

4 in the ....., nodes are expanded in the same order in which they are generated

A Best first
B A*
C BFS
D DFS

5 ....., such as cameras, are true observers of the environment: they capture signals that are generated by other sources in the environment.

A rigid sensors
B Active sensors
C Passive sensors
D mobile sensors

6 In ..... the solution will be the best possible combination of elements of some given set, such as in the scheduling of jobs in a factory

A optimization
B combinational optimization
C adaptation
D selection

7 ..... involve building a possible solution step by step, at each step picking the most possible candidate from the set of possibilities

A Local search algorithms
B swarm intelligence
C adaptation
D Construction algorithms

8 ..... is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through actuators

A A computer
B An Agent
C A machine
D None of the above

9 ..... Means "Is the algorithm guaranteed to find a solution when there is one?"

A Completeness
B Space complexity
C Optimality
D Time complexity

10 ..... just like DFS, except never go deeper than some depth d

A Depth-limited
B A*
C Best first
D BFS

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Part-II: Answer all of the following questions (45 Marks).

Question 2: (15 marks)

A. Discuss the challenges of using genetic algorithms to solve the Traveling Salesman Problem (TSP). [5 marks]

B. A tree can be defined by two vectors. The first vector is the parents vector -s. The second vector is the children vector -t. Each entry in t matrix is a child to the corresponding parent in the same index in the s matrix. Assuming s, t in order for a given tree as:

s	A	A	B	C	C	C	D	E	E	E	E	G
t	B	C	D	E	F	G	H	I	J	K	L	M

Given the aforementioned tree, apply BFS to traverse the tree showing all agenda content and traced paths until reaching the goal state if exist assuming that the goal state is K

Question 3: (15 marks)

A. Explain the concept of "relaxed problems" and how they are used to design heuristics for solving complex problems.

B. Having the following XO sequence of states a long with their values

- - - X O - X - O	0.500
- - X X O - X - O	0.585
- O X X O - X - O	0.523
- O X X O X X - O	0.569
O O X X O X X - O	0.000

Assume a learning rate (α) of 0.56 and discount factor (γ) of 0.82, what will be updated values adopting TD-based state value update with each move. Assume all rewards are -1 except for the actions leading to the goal state with respect to X-player.

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Question 4: (15 marks)

A. Compare and contrast the capabilities and limitations of manipulators and mobile manipulators. [5 marks]

B. A tree can be defined by two vectors. The first vector is the parents vector -s. The second vector is the children vector -t. Each entry in t matrix is a child to the corresponding parent in the same index in the s matrix. Assume s, t in order, for a given tree as:

s	A	A	A	C	C	C	D	D	F	F	G	H	H	I	I
t	32	C	D	41	F	G	H	I	25	2	20	17	38	45	21

Given that tree, use α-β pruning minimax algorithm to evaluate each node in the tree.

End of questions



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